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Article

Increased Serum Netrin-1 Levels Among Children Diagnosed With ADHD

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Abstract

Objective: ADHD is a common neurodevelopmental disorder of childhood characterized by altered projections of dopaminergic neurons and connectivity issues in various brain regions. In our study, we aim to investigate the potential effects of axon guidance molecules, Netrin-1 and Semaphorins 3A, 4D, and 7A, on these connectivity problems by examining the levels of these molecules in the peripheral blood of children with ADHD compared to healthy controls.

Methods: A total of 43 children with ADHD and 40 healthy controls 6 to 12 years of age were included in the study. The K-SADS-PL was administered to exclude any additional psychopathologies (excluding ODD in the ADHD group). The Revised Child Anxiety and Depression Scale was provided to the children, while parents completed the revised Conners' Parent Rating Scale (CPRS-R), the Social Responsiveness Scale (SRS), and the Behavior Rating Inventory for Executive Functioning (BRIEF). Additionally, teachers of the ADHD group were given the Conners' Teacher Rating Scale. Furthermore, the WISC-4 was administered to assess the IQ profile of 35 children in the ADHD group.

Results: Netrin-1 was found to be statistically higher in the ADHD group. When the R-CADS scores were recalculated taking the covariate into account, the significant increase in the ADHD group remained. There was no statistically significant difference between the two groups for semaphorin 3A, 4D, and 7A. The positive correlation of Netrin-1 with the hyperactivity subscores on the CPRS-R and the Global Executive Score on the BRIEF scale is noteworthy.

Conclusion: Netrin-1 may play a role in the etiopathogenesis of ADHD. Further studies are needed to clarify the relationship between Netrin-1 and ADHD. (*J. of Att. Dis. 2026; 30(6) 765-772*)

Keywords

ADHD, brain plasticity, children, diagnosis, mental health

Introduction

ADHD is one of the most common neurodevelopmental disorders of childhood characterized by hyperactivity, inattention, and impulsivity (Sharma & Couture, 2013). According to DSM-5 diagnostic criteria, the prevalence of ADHD is 11.3% in children under 12 years of age and 12.7% in adolescents aged 12 to 18 years (Salari et al., 2023). In a study conducted in Turkey, the prevalence of ADHD among primary school children was 12.7% (Ercan et al., 2015).

Its etiopathogenesis has been reported to involve dopaminergic dysfunction, particularly in the mesocortical, mesolimbic, and nigrostriatal pathways (Del Campo et al., 2011; Sonuga-Barke, 2005). It is thought that dysfunction in the mesocortical dopamine pathway is associated with cognitive impairment, and that hypoactivity in the mesolimbic dopaminergic pathway plays a role in the motivational impairment seen in ADHD patients. The mesolimbic pathway is also an important part of the "reward" circuitry that is impaired in ADHD (Mehta et al., 2019). At the same time, the striatum is the largest source of dopamine in the brain, suggesting an important role for dopamine in attention. Connections to the striatum are a major source of dopaminergic synapses and have been suggested to play a role in ADHD (Dougherty et al., 1999).

ADHD is increasingly seen as a syndrome of dysconnectivity rather than a disorder defined by changes in specific brain regions (Cortese et al., 2021). Study of functional connectivity in ADHD suggests that dopaminergic functional connectivity is a potential prognostic factor in ADHD (Zaher et al., 2024). It has also been demonstrated that ADHD patients exhibit significantly greater connectivity

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variability in the cingulo-temporal, cingulo-parietal, fronto-temporal, and fronto-parietal networks compared to typically developing children (Misra & Gandhi, 2023).

Netrin and Semaphorins are molecules that regulate the connectivity of neurons in the brain, have important roles in axon guidance and therefore may have an effect on the etiology of ADHD. Netrin was first discovered in the 1990s as a vertebrate homologue of the laminin-associated UNC-6 protein found in Caenorhabditis elegans (Serafini et al., 1994). Netrins are categorized into two families of receptors: The DCC (deleted in colorectal cancer) family, which includes DCC and neogenin, and the UNC5 family, which includes Unc5a, Unc5b, Unc5c, and Unc5d proteins (Engelkamp, 2002; Leonardo et al., 1997; Livesey, 1999).

In the central nervous system, netrins influence cell and axon migration and have been demonstrated to facilitate axon arborization and synapse formation during neural development (K. L. W. Sun et al., 2011). Additionally, they are known to promote cortical axon branching (Dent et al., 2004). Beyond axon guidance, netrins are involved in synaptic plasticity, synaptogenesis, cell migration, cell survival, and axon regeneration (Boyer & Gupton, 2018; Glasgow et al., 2021; Kefeli et al., 2017; K. L. W. Sun et al., 2011). Studies indicate that netrin-1 may be implicated in the function, plasticity, and peripubertal reorganization of mesocorticolimbic dopaminergic systems (Manitt et al., 2010). In animal models, the presence of netrin-1 has been shown to restore dopaminergic axonal projections in the striatum, whereas its absence is linked to dopaminergic cell loss in the substantia nigra (Jasmin et al., 2021). Moreover, netrin-1 has been found to maintain the integrity of the blood-brain barrier, mitigate neuroinflammation by reducing chemokine secretion from central nervous system endothelial cells, and exert an anti-inflammatory effect on the blood-brain barrier in response to cytokine-induced disruptions (Podjaski et al., 2015).

Semaphorins (Sema) constitute a large, evolutionarily conserved family of proteins that function as both secreted and membrane-associated guidance cues (Alto & Terman, 2017). Research has demonstrated the essential role of semaphorin signaling in accurately directing specific subsets of afferent projections within the nervous system to their designated targets (Tran et al., 2007). Other functions in the central nervous system include facilitating synapse formation (Inoue et al., 2018; McDermott et al., 2018), contributing to homeostatic synaptic plasticity (Orr et al., 2017), promoting axon terminal branching (Cioni et al., 2013; Jung et al., 2019), enabling axonal pruning (Liu et al., 2005), influencing dendritic morphogenesis and arborization (Danelon et al., 2020; Yamashita et al., 2014), and aiding in the removal of ectopic synapses (Mohan et al., 2019; Tran et al., 2009).

Among the various semaphorins, Semaphorin 3A has been extensively studied, revealing numerous functions within the central nervous system. The interactions between Semaphorin 3A and neuropilin are pivotal for the migration of striatal and cortical interneurons (Marín et al., 2001). Furthermore, Semaphorin 3A has been associated with the inhibition of neurogenesis in the subventricular zone (T. Sun et al., 2016) and has implications in neuroprotection during neural stress (Majed et al., 2006). Sema4D is a biologically active molecule in both its membrane-associated and soluble forms (Smith et al., 2015) and is expressed by oligodendrocytes by oligodendrocytes, where it plays a role in inhibiting axon regeneration (Moreau-Fauvarque et al., 2003). Semaphorin 7A is unique as the only glycosylphosphatidylinositol (GPI)-anchored semaphorin identified to date, expressed in both glial and neuronal cells in the adult central nervous system (Xu et al., 1998). It is suggested to facilitate not only axon guidance and neuronal migration but also the inhibition of neurogenesis in the hippocampus (Jongbloets et al., 2017; Pasterkamp et al., 2007). Additional roles of Semaphorin 7A include involvement in glial scar formation and maturation following spinal cord injury (Kopp et al., 2010) as well as contributions to gliosis (Gutiérrez-Franco et al., 2016). These various functions suggest a potential role for Semaphorin 7A in neurodegenerative processes (Gutiérrez-Franco et al., 2016).

The objective of this study is to explore the potential roles of netrin and semaphorin guidance molecules in the etiopathogenesis of ADHD, with a specific emphasis on their impact on the development of dopaminergic pathways and neural connectivity. The primary hypothesis posits that dysregulation of netrin-1 and semaphorin signaling contributes to the abnormal formation of dopaminergic circuits and synaptic plasticity that underlie the cognitive, motivational, and behavioral deficits observed in individuals with ADHD. Given the connectivity deficits associated with ADHD, our study will examine the blood levels of netrin-1, an important factor in dopamine axon guidance, as well as semaphorin 3A, 4D, and 7A, to investigate their roles in the etiopathogenesis of ADHD.

Materials and Methods

Participants

Children aged 6 to 12 years who were admitted to the outpatient clinics of Ankara University Faculty of Medicine, Department of Child and Adolescent Psychiatry, between June 2023 and February 2024 were included in this study. Informed consent was obtained from all children and their families who participated. The ADHD group comprised 48 patients who had recently received a diagnosis of ADHD and did not have any comorbid psychiatric disorders, aside from oppositional defiant disorder (ODD). In contrast, the control group consisted of 42 healthy participants who did not meet any psychiatric diagnostic criteria. The exclusion criteria for both groups included: known systemic illnesses, presence of any psychiatric diagnosis (aside from ODD in

the ADHD group), regular medication use, infection-related illnesses within the past month, and illiteracy. Approval from the Ethics Committee at Ankara University Faculty of Medicine was secured (11.05.2023/I05-291-23).

A total of 133 children were invited to participate in the study, including 56 for the control group and 77 for the ADHD group. In the control group, 14 children declined participation and were subsequently excluded from the study. In the ADHD group, 18 children either refused to participate or withdrew during the study; these participants were also excluded. Additionally, 11 children who provided blood samples but did not complete the Wechsler Intelligence Scale for Children Fourth Edition (WISC-4) were excluded from further analysis. Ultimately, 42 participants were included in the healthy control (HC) group, while 48 participants were included in the ADHD group, all aged between 6 and 12 years. Seven children (five from the ADHD group and two from the HC group) were excluded from the analysis due to missing blood values.

Procedure

The assessment of the children was conducted by the researcher using the Kiddie Schedule for Affective Disorders and Schizophrenia for School-Age Children-Present and Lifetime Version (K-SADS-PL-DSM-5-Turkish). The parents of all participants completed the Revised Conners' Parent Rating Scale (CPRS-R), the Social Responsiveness Scale (SRS), and the Behavior Rating Inventory for Executive Functioning (BRIEF) parent report. Additionally, teachers of children in the ADHD group completed the Conners' Teacher Rating Scale (CTRS).

All participants were administered Turkish version of the Revised Child Anxiety and Depression Scale (RCADS) to screen for anxiety and depression scores. Additionally, the intelligence quotient (IQ) of children in the ADHD group was evaluated using the Wechsler Intelligence Scale for Children Fourth Edition (WISC-4).

All participants underwent screening for anxiety and depression using the Turkish version of the Revised Child Anxiety and Depression Scale (RCADS). Furthermore, the intelligence quotient (IQ) of children in the ADHD group was assessed using the WISC-4. The CPRS-R is a widely utilized tool for evaluating ADHD and related behavioral concerns in children and adolescents, offering insights into inattention, hyperactivity, and oppositional behaviors based on reports from parents, teachers, or self-reports (Conners et al., 1998). The BRIEF is a standardized assessment tool that evaluates executive function in children and adolescents, measuring aspects such as working memory, emotional regulation, and cognitive flexibility through reports from parents, teachers, or self-reports (Baron, 2000; Gioia et al., 2000). The RCADS is a self-report questionnaire employed to assess symptoms of various anxiety disorders and major depressive disorder based on DSM criteria (Chorpita et al., 2000). The SRS is a standardized measure that evaluates the severity of social impairments associated with autism spectrum disorders by assessing social communication, interaction, and restrictive behaviors across multiple contexts (Constantino et al., 2003).

Biochemical Examination

For laboratory analysis, blood specimens were collected from all participants at 8:00 and 10:30 AM after a fasting period of at least eight hours. Blood samples collected in biochemistry tubes were centrifuged at 4000 × *g* for 10 min; the resultant sera were stored in Eppendorf tubes at −80°C until analysis. Serum samples were analyzed in the research laboratory of the Department of Medical Biochemistry at Ankara University, utilizing ELISA kits for these markers. The levels of Netrin-1 (Ntn1), Semaphorin 3A (SEMA3A), Semaphorin 4D (SEMA4D), and Semaphorin 7A (SEMA7A) in the serum samples were assessed using a sandwich ELISA method.

The blood neutrophil-lymphocyte ratio (NLR) was also evaluated as an inflammatory parameter, derived from complete blood counts. While age, gender, and measurement methods may influence NLR values, a serum NLR ratio of approximately 0.7 to 3.5 is considered within the normal range (Forget et al., 2017). Therefore, children with an NLR ratio exceeding 3.5 were excluded from the study.

Statistical Analysis

Given the absence of comparable studies in the literature, the effect sizes proposed by Cohen (1988) were utilized to determine the appropriate sample size (Cohen, 1988). Assuming an effect size of 0.8, a type I error rate of 0.05, and a statistical power of 95%, the required sample size was calculated to be 84 participants, with 42 in each group (Cohen, 2013).

Statistical analyses were performed using the SPSS 25.0 software package. Descriptive analyses were conducted to examine demographic data, and the relationship between categorical variables between the two groups was analyzed using the chi-square test. Comparisons of biomarker levels and scale scores between the two groups were assessed using the *t*-test if the data met the assumptions of normal distribution; otherwise, the Mann-Whitney *U* test was employed. The relationship between scale scores and serum levels of biomarkers was analyzed through correlation analysis. To account for potential psychiatric confounders affecting biomarker levels between the two groups, anxiety and depression scores were included as covariates in an ANCOVA analysis. A *p*-value of less than .05 was considered statistically significant for all analyses.

Results

A total of 43 participants were included in the ADHD group (comprising 25 boys and 18 girls) and 40 participants in the healthy control group (including 22 boys and 18 girls). The mean age of participants in the ADHD group was 8.34 years ($SD = 1.53$), while the mean age of the control group was 8.85 years ($SD = 1.64$). No statistically significant differences were observed between the groups regarding gender ($p = .773$) or age ($p = .150$). Demographic data for the participants are presented in Table 1.

When comparing the R-CADS subscale scores between groups, no significant difference was found for the Total Anxiety score ($p = .096$). However, the Depression score ($p = .004$) and the Total Internalizing score ($p = .039$) were significantly higher in the ADHD group. Additionally, all subscale scores of the BRIEF were significantly elevated in the ADHD group compared to healthy controls ($p < .001$), and the total score of the SRS was also significantly higher in the ADHD group ($p < .001$). Among the CPRS-R subscale scores, all but the Perfectionism score ($p = .140$) were significantly higher in the ADHD group compared to the healthy control group, including scores for Psychosomatic ($p = .003$) and other subscales ($p < .001$).

Serum Netrin-1 levels were significantly higher in the ADHD group compared to the control group ($p = .016$; refer to Table 2). Even when recalculated with covariates, the significant elevation of Netrin-1 levels in the ADHD group remained evident ($f = 5.392$, $p = .023$). Conversely, no significant differences were found between the two groups for serum levels of Semaphorin 3A, 4D, or 7A ($p = .446$, .099, and .676, respectively; Table 2). The serum NLR ratios are also summarized in Table 2.

Positive correlations were found between serum Netrin-1 levels and levels of Semaphorin 3A, 4D, and 7A (in the ADHD group: Sema3A, 4D, and 7A $p < .001$; in the healthy control group: Sema3A and 4D $p < .001$, Sema7A $p = .004$). Additionally, serum Netrin-1 levels were positively correlated with the CPRS-R subscores: Hyperactivity ($r = .512$, $p = .001$), DSM-4 Hyperactivity-Impulsivity ($r = .396$, $p = .017$), and DSM-4 Total score ($r = .353$, $p = .037$) in the ADHD group.

Moreover, serum Netrin-1 levels were positively associated with various subscale scores of the BRIEF in the ADHD group, including the Inhibit score ($r = 0.408$, $p = .007$), Emotional Control score ($r = .407$, $p = .007$), Working Memory score ($r = .371$, $p = .014$), Organization of Materials score ($r = .330$, $p = .031$), Monitor score ($r = .315$, $p = .040$), Behavior Regulation Index score ($r = .391$, $p = .010$), Metacognition Index score ($r = .334$, $p = .029$), and Global Executive Score ($r = .367$, $p = .015$) .

Additionally, a significant positive correlation was observed between serum Netrin-1 levels and the total score of the SRS in the ADHD group ($p = .040$).

Table 1. Demographic Data of the Participants.

	ADHD		Control		
Variable	$n = 43$	%	$n = 40$	%	p
Gender					
Girl	18	41.9	18	45	$p = .773$
Boy	25	58.1	22	55	
	Mean ± SD		Mean ± SD		p
Age	8.34 ± 1.53		8.85 ± 1.642		$p = .15$

Note. Chi-square test, Independent Sample T test. SD = standard deviation.

Discussion

In our study, we found that levels of the axon guidance molecule Netrin-1 were significantly elevated in children diagnosed with ADHD. In contrast, we observed no significant differences in the levels of other axon guidance molecules, namely Semaphorin 3A, 4D, and 7A.

Netrin-1 is a key axon guidance molecule that plays a crucial role in neuron migration and in determining the trajectory of growing axons and dendrites during the embryonic development of the nervous system (Huber et al., 2003). It is particularly important for the development of dopaminergic neurons extending to the striatum, continuing through puberty, as well as for the proper formation of the mesocorticolimbic pathway (Hoops & Flores, 2017). Dysfunction in axon guidance molecules has been implicated in various neurological disorders (Wang et al., 2022).

Alterations in brain connectivity are also documented in neurodegenerative diseases such as Alzheimer's and Parkinson's disease (Conti et al., 2022; Y. Sun et al., 2014). Notably, a study found that serum Netrin-1 levels in patients with Alzheimer's disease were significantly lower compared to healthy controls (Zhang et al., 2021). Moreover, research by Ahn et al. (2020) in a Parkinson's animal model demonstrated that processes leading to decreased Netrin-1 levels may induce apoptosis in dopaminergic neurons. The findings of our study, which indicate elevated Netrin-1 levels, contrast with reports from Alzheimer's and Parkinson's disease research. This discrepancy may be attributable to the inclusion of younger participants in our study.

Previous studies suggest that the striatal and cerebellar regions are capable of modulating behavior, refined through experiential learning, during childhood development. Such modulation may occur via afferent and efferent pathways to the prefrontal cortex (Amso & Casey, 2006; Nigg & Casey, 2005). Furthermore, research by Nigg and Casey (2005). suggests that these pathways might be less robust in children diagnosed with ADHD, potentially affecting their behavioral regulation. Costa Dias et al. (2013). provided evidence that the nucleus accumbens (NAcc) exhibits atypical connectivity with various cortical regions in children with ADHD. Additionally, these regions are associated with

Table 2. Comparison of The Serum Levels of Markers Between Groups.

	ADHD		Control			
	Mean	*SD*	Mean	*SD*	*t*	*p*
Netrin-1	471.010	318.346	328.21	197.185	−2.475	.016*
Semaphorin 3A	13.971	7.172	13.00	3.998	−0.766	.446
Semaphorin 4D	7.332	5.049	5.730	3.626	−1.668	.099
Semaphorin 7A	13.411	9.098	12.706	5.712	−0.419	.676
Neutrophil/ lymphocyte ratio	1.430	0.868	1.331	0.526	−0.614	.541

Note. Comparison of serum Netrin-1 levels with controls using Independent Sample *T* test. Netrin-1 pg/ml, Semaphorin 3A-4D-7A ng/ml. *SD* = standard deviation.
*$p < .05$.

abnormal patterns of spontaneous neural activity in individuals diagnosed with ADHD. The abnormal regional connections and activity patterns observed in ADHD may result from increased Netrin-1 levels leading to excessive neuronal connections. In essence, while Netrin-1 facilitates the guidance of altered dopaminergic neurons in ADHD, it may also induce aberrant axonal connections through various mechanisms. The increased plasma levels of Netrin-1 observed in our study may represent a compensatory response to impaired axon connectivity.

Moreover, we found that Netrin-1 levels were positively correlated with all BRIEF subscales, except for Plan/ Organize, Initiate, and Set Shifting. The brain regions involved in executive functions include the prefrontal cortex, striatum, and hippocampus. Netrin-1 may influence the connectivity of prefrontal neurons through the DCC receptor, with potential effects on executive functions such as working memory. Additionally, Netrin-1 may enhance memory processes by increasing synaptic plasticity of hippocampal neurons (Glasgow et al., 2021). While lower levels of Netrin-1 are associated with impaired memory and hippocampal function (López-Huerta et al., 2013), our findings suggest that elevated Netrin-1 correlates with executive function impairments. This indicates that maintaining Netrin-1 levels within an optimal range is essential for proper neural remodeling. Furthermore, our results indicated that serum Netrin-1 levels were positively correlated with hyperactivity and DSM-IV total scores on the CPRS-R. These findings underscore the significance of Netrin-1 in brain regions, such as the prefrontal cortex and NAcc, which are integral to behaviors associated with hyperactivity in ADHD.

ADHD is a neurodevelopmental disorder with an incomplete understanding of its etiopathogenesis, which remains an area of active investigation. In our study, we propose that axon guidance molecules, previously overlooked in the context of neurodevelopmental disorders, may contribute to the etiology of ADHD and warrant further research. Currently, there are no biomarkers available for diagnosis or treatment monitoring in neurodevelopmental disorders. Thus, we sought to identify a biomarker that could assist clinicians in diagnosis and treatment. Axon guidance molecules, particularly Netrin-1, may be valuable in this regard. Future studies may enhance clinical practice by comparing plasma levels of Netrin-1 with treatment responses, especially as changes in Netrin-1 within the NAcc can influence the efficacy of stimulant medications.

The strengths of our study include its status as the first investigation of axon guidance molecules in children with ADHD within the existing literature and the exclusion of additional clinical and psychiatric diagnoses in ADHD participants. However, our study has certain limitations, including its conduct at a single center with a clinical sample derived from outpatient settings, as well as the assessment of these molecules using peripheral blood samples. Additionally, our reliance on parent-report questionnaires without incorporating cognitive tests presents another limitation. In the analysis phase of our study, we did not observe any significant differences between groups for non-netrin markers. Therefore, to avoid the false discovery rate, we did not perform further analyses on these markers. Thus, there is a compelling need for further studies with larger and more diverse samples to validate and expand upon our findings

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Ethical Considerations

This study approved by Ankara University Faculty of Medicine Ethical Committee (approval date: 11.05.2023 approval number: İ05-291-23).

Consent to Participate

Informed consent was obtained from all participant.

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Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

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